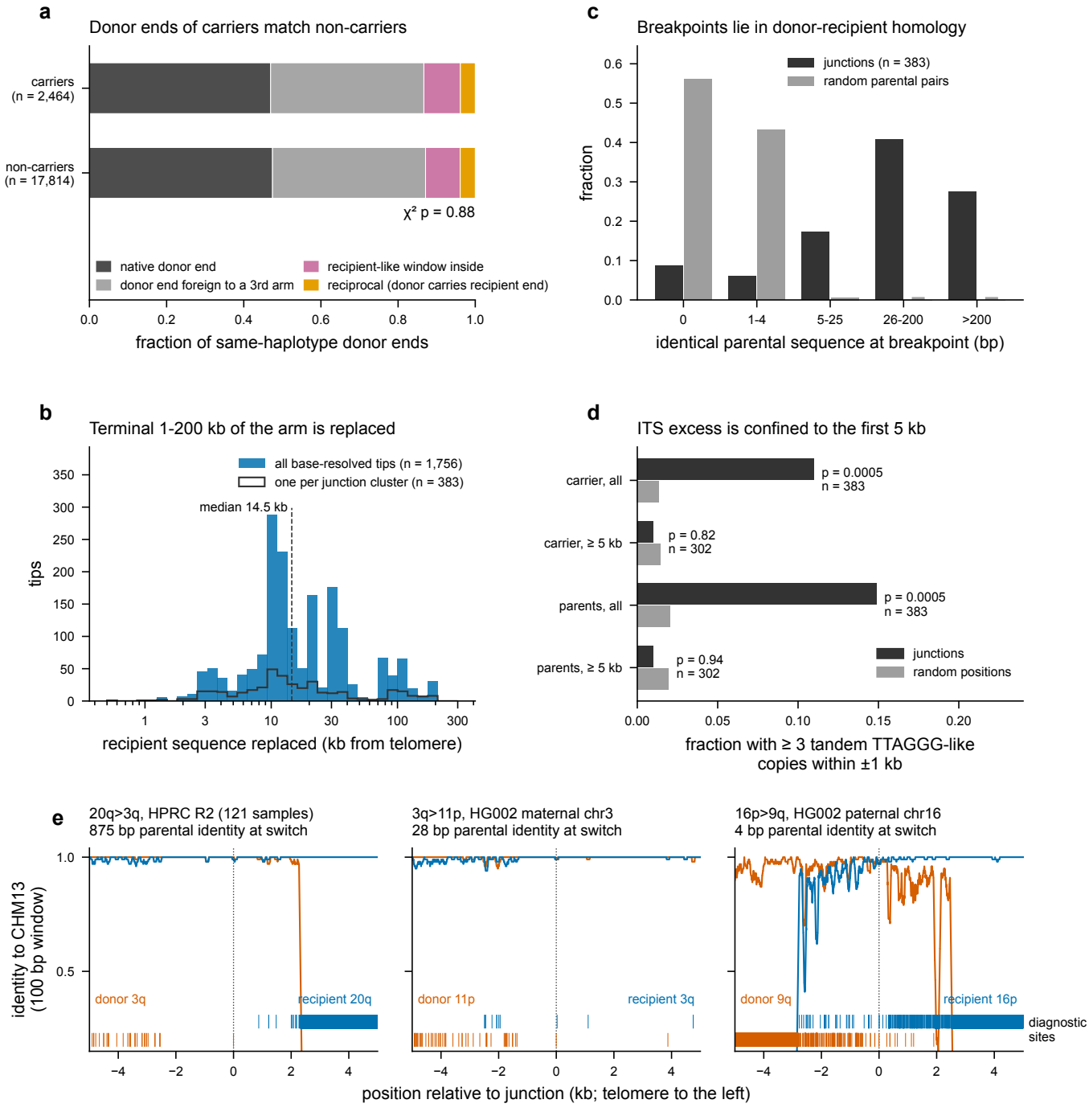

a
Donor ends of carriers match non-carriers
carriers (n = 2,464)
non-carriers (n = 17,814)
χ² p = 0.88
native donor end
donor end foreign to a 3rd arm
recipient-like window inside
reciprocal (donor carries recipient end)
0.0
0.2
0.4
0.6
0.8
1.0
fraction of same-haplotype donor ends
c
Breakpoints lie in donor-recipient homology
junctions (n = 383)
random parental pairs
fraction
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
1-4
5-25
26-200
>200
identical parental sequence at breakpoint (bp)
b
Terminal 1-200 kb of the arm is replaced
all base-resolved tips (n = 1,756)
one per junction cluster (n = 383)
median 14.5 kb
tips
0
50
100
150
200
250
300
350
1
3
10
30
100
300
recipient sequence replaced (kb from telomere)
d
ITS excess is confined to the first 5 kb
carrier, all
p = 0.0005
n = 383
carrier, ≥ 5 kb
p = 0.82
n = 302
parents, all
p = 0.0005
n = 383
parents, ≥ 5 kb
p = 0.94
n = 302
junctions
random positions
0.00
0.05
0.10
0.15
0.20
fraction with ≥ 3 tandem TTAGGG-like copies within ±1 kb
e
20q>3q, HPRC R2 (121 samples)
875 bp parental identity at switch
3q>11p, HG002 maternal chr3
28 bp parental identity at switch
16p>9q, HG002 paternal chr16
4 bp parental identity at switch
identity to CHM13 (100 bp window)
1.0
0.5
donor 3q
recipient 20q
donor 11p
recipient 3q
donor 9q
recipient 16p
diagnostic sites
−4
−2
0
2
4
position relative to junction (kb; telomere to the left)